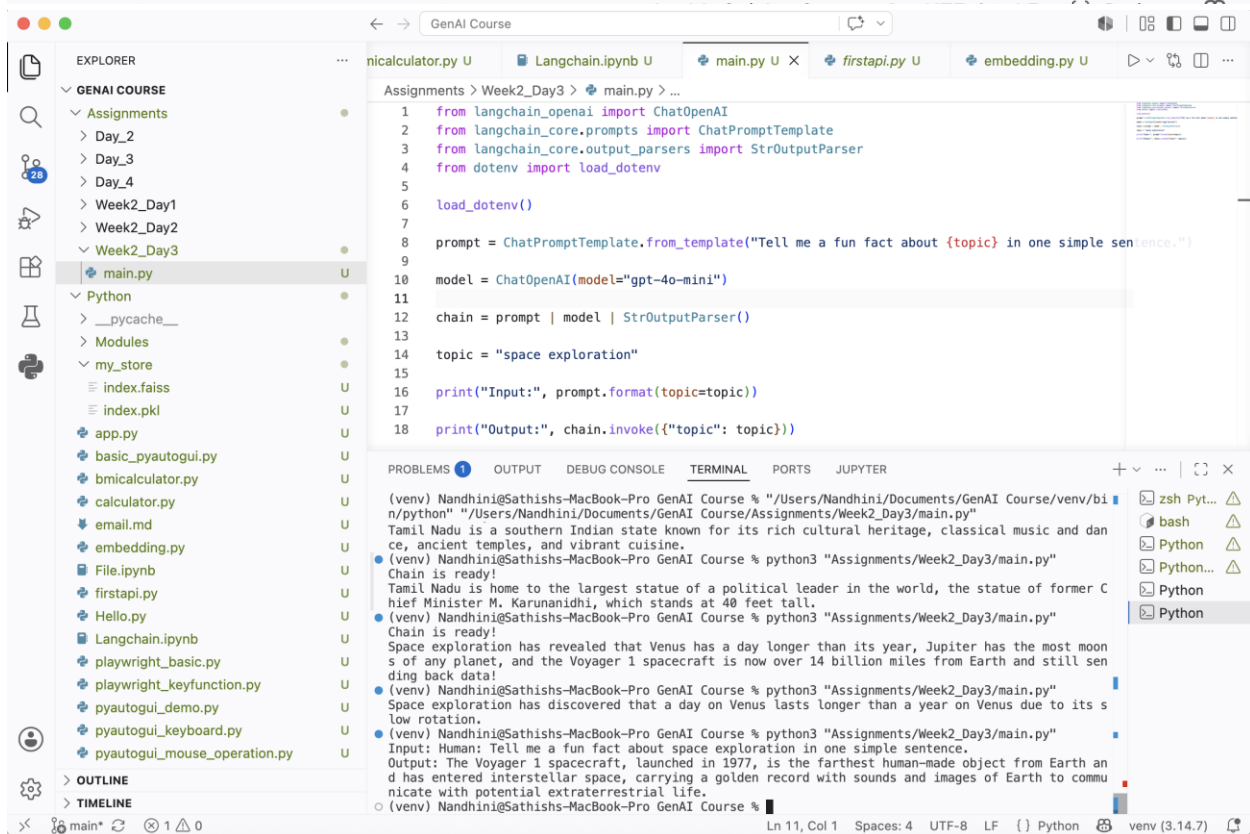


Langchain Assignment

- (venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % python3 "Assignments/Week2_Day3/main.py"
Input: Human: Tell me a fun fact about space exploration in one simple sentence.
Output: The Voyager 1 spacecraft, launched in 1977, is the farthest human-made object from Earth and has entered interstellar space, carrying a golden record with sounds and images of Earth to communicate with potential extraterrestrial life.
- (venv) Nandhini@Sathishs-MacBook-Pro GenAI Course %



```
EXPLORER
  GENAI COURSE
    Assignments
      Day_2
      Day_3
      Day_4
      Week2_Day1
      Week2_Day2
      Week2_Day3
        main.py
    Python
      __pycache__
      Modules
      my_store
        index.faiss
        index.pkl
      app.py
      basic_pyautogui.py
      bmiccalculator.py
      calculator.py
      email.md
      embedding.py
      File.ipynb
      firstapi.py
      Hello.py
      Langchain.ipynb
      playwright_basic.py
      playwright_keyfunction.py
      pyautogui_demo.py
      pyautogui_keyboard.py
      pyautogui_mouse_operation.py
    OUTLINE
    TIMELINE

main.py
1 from langchain_openai import ChatOpenAI
2 from langchain_core.prompts import ChatPromptTemplate
3 from langchain_core.output_parsers import StrOutputParser
4 from dotenv import load_dotenv
5
6 load_dotenv()
7
8 prompt = ChatPromptTemplate.from_template("Tell me a fun fact about {topic} in one simple sentence.")
9
10 model = ChatOpenAI(model="gpt-4o-mini")
11
12 chain = prompt | model | StrOutputParser()
13
14 topic = "space exploration"
15
16 print("Input:", prompt.format(topic=topic))
17
18 print("Output:", chain.invoke({"topic": topic}))

TERMINAL
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % "/Users/Nandhini/Documents/GenAI Course/venv/bin/python3" "/Users/Nandhini/Documents/GenAI Course/Assignments/Week2_Day3/main.py"
Tamil Nadu is a southern Indian state known for its rich cultural heritage, classical music and dance, ancient temples, and vibrant cuisine.
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % python3 "Assignments/Week2_Day3/main.py"
Chain is ready!
Tamil Nadu is home to the largest statue of a political leader in the world, the statue of former Chief Minister M. Karunanidhi, which stands at 40 feet tall.
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % python3 "Assignments/Week2_Day3/main.py"
Chain is ready!
Space exploration has revealed that Venus has a day longer than its year, Jupiter has the most moons of any planet, and the Voyager 1 spacecraft is now over 14 billion miles from Earth and still sending back data!
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % python3 "Assignments/Week2_Day3/main.py"
Space exploration has discovered that a day on Venus lasts longer than a year on Venus due to its slow rotation.
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course % python3 "Assignments/Week2_Day3/main.py"
Input: Human: Tell me a fun fact about space exploration in one simple sentence.
Output: The Voyager 1 spacecraft, launched in 1977, is the farthest human-made object from Earth and has entered interstellar space, carrying a golden record with sounds and images of Earth to communicate with potential extraterrestrial life.
(venv) Nandhini@Sathishs-MacBook-Pro GenAI Course %
```

LangSmith

Application

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Datasets & Experiments

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Prompts

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Studio

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my-first-appID

Retention14dDashboardNew

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Add filter

☐	☒	Name	Input	Output	Error	Start Time	Latency	Data
☐	☒	RunnableSequence	space exploration	The Voyager 1 spacecraft...		9/2/2026, 10:1...	1.68s	
☐	☒	RunnableSequence	space exploration	Space exploration has dis...		9/2/2026, 10:0...	1.86s	
☐	☒	RunnableSequence	space exploration	Space exploration has rev...		9/2/2026, 9:56...	1.28s	
☐	☒	RunnableSequence	Tamil Nadu	Tamil Nadu is home to the...		9/2/2026, 9:55...	1.89s	
☐	☒	RunnableSequence	Tamil Nadu	Tamil Nadu is a southern I...		9/2/2026, 9:49...	2.50s	

Trace

 RunnableSequence

gpt-4o-mini

1.61s

66

RunnableSequence ID

Feedback	Input	Output	Attributes
----------	-------	--------	------------

Feedback

+ Add feedback

- Input

Markdown ▾

{ } Fields

- topic space exploration

Output

Markdown ▾

Fields

- output

The Voyager 1 spacecraft, launched in 1977, is the farthest human-made object from Earth and has entered interstellar space, carrying a golden record with sounds and images of Earth to communicate with potential extraterrestrial life.

- Attributes